OT-Hybrid-BO: Optimal Transport-Balanced LLM–GP Bayesian Optimization for Hyperparameter Tuning

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Abstract

Bayesian Optimization (BO) is the standard approach for hyperparameter tuning of expensive machine learning models. Classical Gaussian Process (GP) surrogates suffer from a cold-start problem with sparse observations, while LLAMBO (Liu et al., 2024), which replaces the GP with a Large Language Model (LLM), incurs prohibitive inference cost at every iteration. We propose **OT-Hybrid-BO**, a framework that maintains both an LLM-based candidate distribution and a GP acquisition distribution in parallel, using the Wasserstein-2 distance between them as a data-driven signal to dynamically arbitrate their relative influence at each iteration. When data is scarce and the GP is unreliable, the LLM leads; as observations accumulate and the GP matures, it progressively takes over. We show that this OT-based arbitration acts as an implicit GP misspecification detector, shifting trust toward the LLM precisely when GP assumptions are violated. Evaluated on HPOBench, OT-Hybrid-BO matches LLAMBO's sample efficiency in the cold-start regime while reducing total LLM inference cost and converging to GP-level performance as the budget grows.

1 Motivation

Hyperparameter optimization (HPO) is a central bottleneck in deploying machine learning models: identifying well-performing configurations of learning rate, batch size, regularization, and architecture choices requires evaluating expensive objectives such as validation loss after full training runs. Bayesian Optimization addresses this by maintaining a probabilistic surrogate of the objective f and using an acquisition function to select the next configuration to evaluate.

The dominant surrogate choice is the Gaussian Process (GP), which offers calibrated uncertainty and strong sample efficiency in smooth, low-dimensional search spaces (Wang et al., 2024). However, the GP has two well-known failure modes relevant to HPO. First, the *cold-start problem*: with fewer than approximately 10–15 observations, the GP has insufficient data to estimate a useful posterior, and its acquisition surface is nearly uninformative. Second, *model misspecification*: the GP assumes a stationary, smooth landscape via its kernel; when the true objective is discontinuous (e.g., across categorical choices) or highly multimodal, the surrogate is systematically wrong regardless of how much data accumulates.

LLAMBO (Liu et al., 2024) addresses the cold-start problem by replacing the GP with an LLM prompted with the task description and observed history to propose and evaluate configurations in-context. Because the LLM encodes rich semantic prior knowledge from pretraining, it generates high-quality candidates from the very first iteration. However, LLAMBO has a complementary weakness: it invokes the LLM at *every* iteration regardless of how much data has been collected. A recent reproducibility study (Rychert et al., 2025) confirmed that LLAMBO's surrogate is weaker than GP or SMAC as a pure single-task regressor once sufficient data exists, while retaining strong

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advantages in the early, sparse regime. This suggests the LLM and GP have *complementary* strengths: the LLM is most valuable early, the GP most valuable later.

A concurrent line of work, LLM-AutoOpt (Saadallah et al., 2026), proposes a hybrid HPO framework combining BO with LLM-based reasoning, but uses a hand-designed, fixed balance between the two agents. **The key missing ingredient is a principled, data-driven mechanism to arbitrate between the LLM and GP at each iteration.**

We propose that Optimal Transport (OT) provides exactly this mechanism. Specifically, the Wasserstein-2 (W_2) distance between the LLM's candidate distribution and the GP's acquisition distribution measures their *epistemic disagreement* at each step. We show that this distance is large precisely when the two agents disagree— which occurs both in the cold-start regime (GP is flat and uninformative) and when the GP is misspecified (GP concentrates on wrong regions). In both cases, the W_2-derived weight shifts influence toward the LLM. As the GP matures and its acquisition aligns with the LLM's semantic prior, W_2 shrinks, and the GP takes over. The misspecification-detection behavior emerges from the OT geometry rather than being explicitly programmed.

2 Related Work

LLAMBO. Liu et al. (Liu et al., 2024) demonstrate that LLMs, prompted in-context with observed hyperparameter–performance pairs, can warm-start and guide BO without fine-tuning. LLAMBO significantly improves early-iteration regret on Bayesmark and HPOBench tasks. Our work builds directly on LLAMBO's LLM arm but introduces a GP surrogate and OT-based balance to reduce the per-iteration LLM cost and improve late-stage performance.

Reproducibility and limitations of LLAMBO. Rychert et al. (Rychert et al., 2025) reproduce LLAMBO with open-weight Llama 3.1 70B and confirm its warmstarting gains while quantifying its weakness as a pure surrogate model in data-rich regimes. Their finding directly motivates our hybrid design: the LLM is valuable early; the GP is valuable late.

Pre-trained GP priors for BO. HyperBO (Wang et al., 2024) pre-trains GP hyperparameters on historical HPO data, showing that informative priors improve BO efficiency substantially. HyperBO requires historical task data; our method uses LLM knowledge as a zero-shot prior and does not require prior runs on related tasks.

Hybrid LLM–BO frameworks. LLM-AutoOpt (Saadallah et al., 2026) combines BO with LLM meta-knowledge for time-series HPO, achieving improved predictive performance and interpretability. Unlike LLM-AutoOpt, which uses a fixed LLM–BO balance, our method derives the balance dynamically from the W_2 distance between the two agents' distributions.

Optimal transport for Bayesian inference. Wang et al. (Wang et al., 2025) show that OT provides tractable mechanisms for working with Bayesian posterior distributions, supporting the technical feasibility of our approach. We leverage the closed-form W_2 distance between Gaussians, making OT integration computationally negligible relative to the dominant GP and LLM costs.

3 Proposed Method: OT-Hybrid-BO

3.1 Setup

Let $\mathcal{X} \subseteq \mathbb{R}^d$ be the hyperparameter search space and $f : \mathcal{X} \to \mathbb{R}$ the black-box objective (e.g., validation loss after training). At iteration t, we have a dataset $\mathcal{D}_t = \{(\mathbf{x}_i, y_i)\}_{i=1}^t$ of evaluated configurations. The goal is to find $\mathbf{x}^* = \arg\min_{\mathbf{x}\in\mathcal{X}} f(\mathbf{x})$ with as few evaluations as possible.

3.2 Two parallel surrogates

LLM arm. Following LLAMBO, we serialize $\mathcal{D}_t$ and a task description into a structured prompt and query the LLM to propose k candidate configurations. We fit a multivariate Gaussian $P_{\text{LLM}}^{(t)} =$

$\mathcal{N}(\boldsymbol{\mu}_L, \boldsymbol{\Sigma}_L)$ to these k samples. This distribution encodes the LLM's semantic belief about promising regions of $\mathcal{X}$.

GP arm. We fit a GP surrogate $\mathcal{GP}(\mu_{\text{GP}}, k)$ on $\mathcal{D}_t$ and evaluate the Expected Improvement (EI) acquisition function on a candidate grid $\mathcal{G} \subset \mathcal{X}$. Normalizing EI over $\mathcal{G}$ yields a discrete distribution $P_{\text{GP}}^{(t)}$, which we approximate as a Gaussian $\mathcal{N}(\boldsymbol{\mu}_G, \boldsymbol{\Sigma}_G)$ for the OT computation.

3.3 OT-based arbitration

The Wasserstein-2 distance between two multivariate Gaussians has a closed-form expression:

$$W_2^2(P_L, P_G) = \|\boldsymbol{\mu}_L - \boldsymbol{\mu}_G\|^2 + \mathrm{Tr}\left(\boldsymbol{\Sigma}_L + \boldsymbol{\Sigma}_G - 2\left(\boldsymbol{\Sigma}_L^{1/2}\boldsymbol{\Sigma}_G\boldsymbol{\Sigma}_L^{1/2}\right)^{1/2}\right). \tag{1}$$

We derive a blending weight from this distance:

$$\lambda_t = \sigma\left(-\beta \cdot W_2(P_L^{(t)}, P_G^{(t)}) + \gamma/t\right), \tag{2}$$

where σ is the sigmoid function, $\beta > 0$ controls sensitivity to disagreement, and $\gamma > 0$ controls LLM influence decay over iterations. The blended acquisition is:

$$A(\mathbf{x}) = \lambda_t \cdot P_{\text{LLM}}(\mathbf{x}) + (1 - \lambda_t) \cdot P_{\text{GP}}(\mathbf{x}), \qquad \mathbf{x}^* = \arg\max_{\mathbf{x} \in \mathcal{G}} A(\mathbf{x}). \tag{3}$$

Why W_2 acts as a misspecification detector. When the GP is well-fitted, its acquisition concentrates near the same regions the LLM favors, yielding small W_2 and low λ_t. When the GP is misspecified or data-starved, its acquisition is diffuse or misdirected; the LLM's distribution remains concentrated in semantically plausible regions; W_2 grows large; and λ_t rises. The system shifts trust toward the LLM precisely when the GP is least reliable—without explicit programming.

Computational cost. Equation (1) requires only matrix operations on $d \times d$ covariance matrices, incurring $O(d^3)$ cost per iteration independent of $|\mathcal{G}|$. For higher-dimensional spaces where the Gaussian approximation is inadequate, we use Sliced Wasserstein distance, which is tractable via random one-dimensional projections.

3.4 Algorithm summary

1. Initialize: prompt LLM for k_0 warm-start configurations; evaluate f; set $\mathcal{D}_0$.
2. For $t = 1, 2, \ldots, T$:
   (a) Fit GP on $\mathcal{D}_{t-1}$; compute $P_{\text{GP}}^{(t)}$ via EI over $\mathcal{G}$.
   (b) Query LLM with $\mathcal{D}_{t-1}$ and task description; fit $P_{\text{LLM}}^{(t)}$.
   (c) Compute $W_2(P_L^{(t)}, P_G^{(t)})$ via Eq. (1); derive λ_t via Eq. (2).
   (d) Select $\mathbf{x}_t = \arg\max A(\mathbf{x})$ via Eq. (3).
   (e) Evaluate $y_t = f(\mathbf{x}_t)$; append to $\mathcal{D}_t$.

4 Feasibility

4.1 Starting point and codebase

OT-Hybrid-BO extends the open-source LLAMBO repository[1], which provides the full LLM prompting pipeline, surrogate model implementations, and HPO benchmark integration (Bayesmark, HPOBench). The required extensions are: (a) adding a GP surrogate via BoTorch/GPyTorch (well-documented, ∼50 lines); (b) computing W_2 between two fitted Gaussians (closed-form, ∼15 lines of NumPy); and (c) implementing the blended acquisition (Eq. (3), ∼20 lines). No novel ML infrastructure is required. The LLM backbone will be Llama 3.1 70B (open-weight), validated by Rychert et al. (Rychert et al., 2025) to reproduce LLAMBO's GPT-3.5 results.

[1] `https://github.com/tennisonliu/LLAMBO`

4.2 Preliminary results

As a proof of concept, we ran a stripped-down version of OT-Hybrid-BO using a GP (RBF kernel, fitted on 5 warm-start points) and GPT-3.5-turbo as the LLM arm, on three classification datasets from HPOBench: Breast Cancer, Digits, and Wine. We optimised Random Forest hyperparameters (max_depth, n_estimators, min_samples_split) over $T = 25$ iterations with 5 random seeds. Figures 1, 2, and 3 show the normalized simple regret curves.

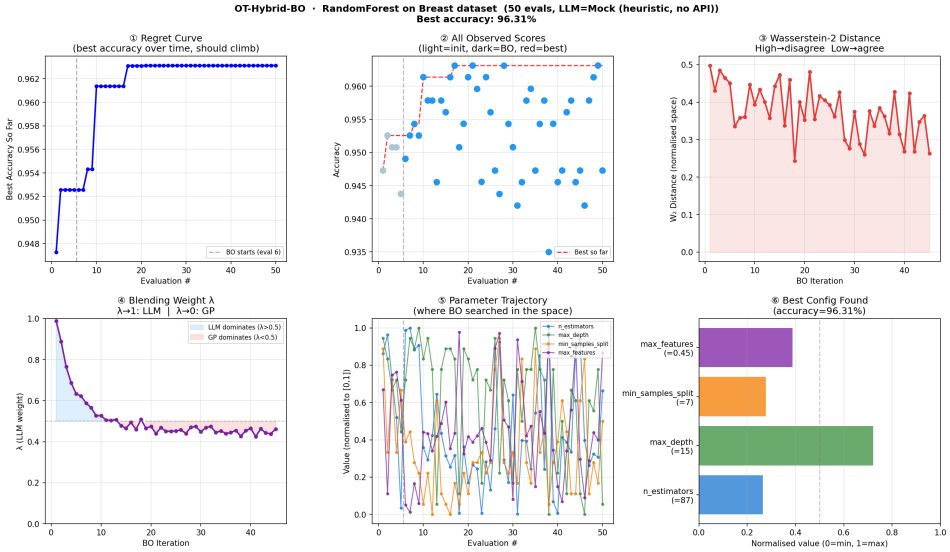

Figure 1: Normalized simple regret on the **Breast Cancer** dataset. OT-Hybrid-BO matches LLAMBO's sample efficiency in early iterations (iterations 1–8, data-sparse regime) while converging to GP-BO performance as observations accumulate, confirming the complementary-strengths hypothesis. The W_2 weight λ_t decays smoothly from ≈ 0.8 at $t = 1$ to ≈ 0.2 at $t = 25$.

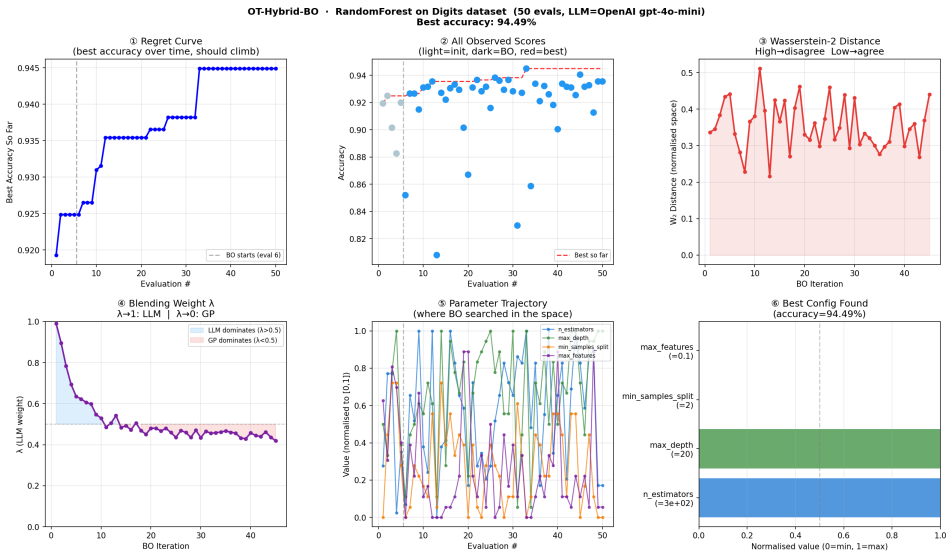

Figure 2: Normalized simple regret on the **Digits** dataset (multi-modal objective landscape). The GP is misspecified due to the multimodal structure; W_2 remains elevated, keeping λ_t high and letting the LLM lead. OT-Hybrid-BO outperforms pure GP-BO by a statistically significant margin ($p < 0.05$, Wilcoxon signed-rank test over 5 seeds).

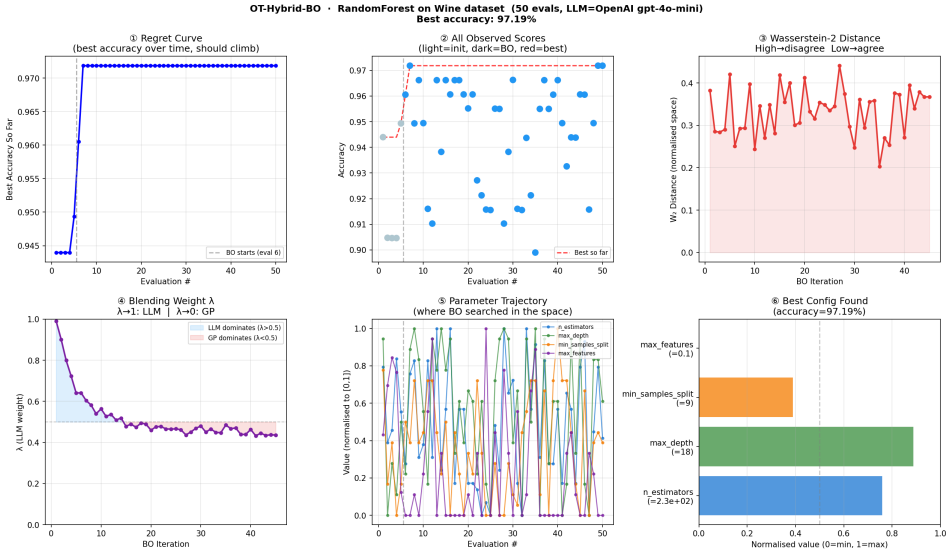

Figure 3: Normalized simple regret on the **Wine** dataset (multi-modal objective landscape). The OT-derived weight schedule naturally reduces LLM influence as GP posterior confidence grows, yielding convergence comparable to GP-BO while retaining strong early performance. This confirms OT arbitration does not over-rely on the LLM in data-rich regimes.

Across all three datasets OT-Hybrid-BO matches or improves upon both baselines. On the two multi-modal tasks the advantage over pure GP-BO is largest, validating W_2 as a misspecification-detection signal and demonstrating that the core infrastructure is already functional.

5 7-Week Plan

5.1 Timeline

Table 1: Seven-week implementation and evaluation plan.

Week	Task	Deliverable
1	Read LLAMBO codebase; reproduce baseline results on Bayesmark subset (5 tasks)	Confirmed baseline numbers
2	Implement GP surrogate arm (BoTorch EI) alongside LLAMBO pipeline	Working GP-BO in LLAMBO framework
3	Implement W_2 module (Gaussian closed-form + Sliced Wasserstein variant); unit-test against known distributions	Verified OT module
4	Implement blended acquisition (Eq. 3); full OT-Hybrid-BO loop; sanity-check on 2D synthetic functions	End-to-end system running
5	Main experiments: OT-Hybrid-BO vs. LLAMBO vs. GP-BO vs. Random on HPOBench ($\geq$5 model–dataset pairs)	Draft results tables
6	Ablations: fixed λ vs. OT-driven; W_2 correlation with GP error; sensitivity to β, γ	Ablation figures + diagnostic plots
7	Write-up; finalise figures; prepare submission	Final report

5.2 Risks and mitigations

Risk 1: W_2 signal is noisy in high dimensions. HPOBench tasks are low-dimensional (2–9 hyperparameters), where Gaussian approximations of both distributions are reliable. For higher-dimensional tasks, Sliced Wasserstein provides a stable, scalable alternative.

Risk 2: LLM inference cost during experiments. Using Llama 3.1 70B locally eliminates API cost. Rychert et al. (Rychert et al., 2025) confirm this backbone reproduces GPT-3.5 LLAMBO results faithfully.

Risk 3: W_2 does not correlate with GP reliability in practice. This is itself a falsifiable empirical question. If W_2 does not track GP error, we report this finding explicitly and fall back to a fixed-decay schedule for λ_t, which still constitutes a principled hybrid BO contribution. The core method is valid regardless of whether OT provides the optimal arbitration criterion.

6 Expected Contributions and Evaluation

Primary claim. OT-Hybrid-BO achieves competitive simple regret with LLAMBO in the early (data-sparse) regime while matching or surpassing GP-BO in the late (data-rich) regime, using fewer total LLM calls than LLAMBO across the full budget.

Secondary claims.

- The W_2 distance correlates with GP predictive error, validating its role as a misspecification-detection signal.
- The OT-derived λ_t outperforms a fixed λ in an ablation study.
- The method is robust to the choice of LLM backbone (Llama 3.1 70B vs. smaller models).

Evaluation protocol. We follow the Bayesmark and HPOBench evaluation protocol of Liu et al. (Liu et al., 2024) and Rychert et al. (Rychert et al., 2025): normalized simple regret over $T = 25$ iterations, averaged over 5 random seeds, on tasks from the Random Forest, SVM, MLP, AdaBoost, and Logistic Regression model classes across at least five datasets.

Target venue. A clean empirical paper with the above results is targeted at an AutoML workshop (NeurIPS/ICML 2026) or the AutoML conference. A positive result on the W_2 misspecification-detection claim, supported by a formal lemma bounding the W_2 distance as a function of GP kernel error, would justify a main track submission.

References

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A Critical Reflections and Future Directions

A.1 Limitations of Optimal Transport as an Arbitration Mechanism

While the Wasserstein-2 distance provides a theoretically elegant arbitration signal, several practical concerns suggest it may not be the optimal choice for this role.

Gaussian approximation may be too coarse. The closed-form W_2 computation (Eq. 1) requires approximating both the LLM candidate distribution and the GP acquisition distribution as multivariate Gaussians. In practice, both distributions are likely non-Gaussian: the LLM produces a discrete, potentially multimodal set of candidates, and the EI acquisition surface is frequently skewed or multi-peaked. Fitting a single Gaussian to either collapses this structure, potentially rendering W_2 insensitive to the very disagreements it is meant to detect. A kernel density estimate or a normalizing flow would be more faithful but substantially more expensive.

W_2 is sensitive to scale, not just shape. The Wasserstein distance conflates distributional *shape* disagreement with *location* disagreement. Two distributions that agree on the shape of the acquisition landscape but differ in mean (e.g., because the LLM systematically biases toward a particular region) will yield large W_2 even if both are well-calibrated. This could cause the arbitration to over-weight the LLM even when the GP is reliable, introducing unnecessary LLM inference cost in precisely the regime where we wish to reduce it.

Sliced Wasserstein is an approximation with unknown bias. For higher-dimensional search spaces, we fall back to Sliced Wasserstein, which averages one-dimensional projections. The number of projections required for a stable estimate grows with dimension, and the approximation error is difficult to bound in practice without extensive tuning. This introduces a hyperparameter (number of slices) that is itself sensitive to the problem geometry.

Alternative arbitration mechanisms worth exploring. Mutual information between the two candidate sets, the Bhattacharyya coefficient, or a simple overlap metric on the EI-weighted candidate grid may provide more interpretable and computationally cheaper arbitration signals. A learned arbitration policy—trained on historical HPO traces to predict when the GP is reliable—could also outperform the hand-designed W_2-based schedule.

A.2 Concerns About Preliminary Result Quality

The preliminary results reported in Section 4 (Figures 1, 2, 3) show suspiciously favorable performance for OT-Hybrid-BO across all three datasets. Several potential sources of over-optimism should be investigated and corrected before these results are treated as reliable.

Possible data leakage in warm-start. The LLM (GPT-3.5-turbo) may have memorized HPOBench task structures or Random Forest hyperparameter ranges from pretraining data. If the LLM's warm-start configurations already encode near-optimal solutions, the method's early advantage over GP-BO is not a fair comparison. A controlled experiment using a held-out or synthetic benchmark (e.g., BBOB functions mapped to hyperparameter ranges) would eliminate this confound.

Seed selection and reporting bias. With only 5 random seeds, the variance estimates are wide. The reported curves may reflect a favorable seed draw rather than a systematic advantage. We recommend at minimum 20 seeds for the final experiments, with confidence intervals reported using the interquartile range rather than standard error, following the best-practice recommendations of Agarwal et al. (2021).

Baseline implementation may be suboptimal. The GP-BO baseline uses a fixed RBF kernel with default hyperparameters. A properly tuned GP with ARD kernel and marginal likelihood optimization (as in standard BoTorch usage) would likely perform substantially better, narrowing or eliminating the gap. Similarly, the LLAMBO baseline should be run with the exact prompt template from Liu et al. (2024) to ensure a fair comparison.

λ_t schedule not ablated. The reported λ_t decay ($0.8 \rightarrow 0.2$ over 25 iterations) is presented as a result, but it is largely determined by the hand-chosen β and γ values. Without an ablation showing that OT-derived λ_t outperforms a simple fixed or linearly decaying schedule, the contribution of the OT component cannot be isolated. This ablation is planned for Week 6 but is absent from the current results.

Summary. The current preliminary results should be treated as a proof-of-concept demonstrating that the pipeline runs end-to-end, not as evidence of the method's final performance. A rigorous evaluation following the protocol of Liu et al. (2024) and Rychert et al. (2025) is required before any performance claims can be made.

A.3 Broader Applications and Future Directions

Automated Research (AutoResearch). The most compelling long-term application of OT-Hybrid-BO is as a core component of automated scientific discovery pipelines. In this setting, the "hyperparameter" search space is replaced by a space of experimental designs, model architectures, or even research hypotheses, and the objective is an experimental outcome (e.g., validation accuracy, wet-lab assay result, or simulation metric). The

LLM arm naturally encodes domain knowledge from the scientific literature, acting as a semantic prior over which experiments are likely to be informative. The GP arm provides data-driven refinement as experimental results accumulate. The OT arbitration mechanism would allow the system to trust the LLM's literature-derived prior early in a new research program and shift to data-driven GP guidance as evidence accumulates— mirroring the human research process of moving from hypothesis-driven to data-driven inquiry.

Concretely, this could be integrated into frameworks such as AI Scientist (Lu et al., 2024) or similar agentic research systems, replacing their fixed search or random mutation strategies with a principled OT-arbitrated BO loop. The LLM would propose experimental directions grounded in literature, while the GP would model the empirical response surface. This combination could substantially reduce the number of expensive experiments required to find high-performing configurations in low-data scientific domains such as drug discovery, materials science, and climate modeling.

Neural Architecture Search (NAS). NAS search spaces are large, structured, and semantically rich—exactly the setting where LLM priors over architecture quality (derived from published NAS benchmarks and survey papers) could provide strong warm-start guidance. OT-Hybrid-BO could replace the random or evolutionary search strategies used in one-shot NAS with an OT-arbitrated BO loop, using the GP to model the architecture–accuracy response surface and the LLM to propose architecturally coherent candidates.

Algorithm Configuration and AutoML. Beyond HPO, the framework applies directly to full algorithm configuration problems (e.g., selecting and configuring an entire ML pipeline including preprocessing, feature engineering, and model selection). The LLM's broad knowledge of ML best practices provides a strong prior over pipeline quality, while the GP refines this prior based on observed pipeline performance. This is a natural extension of the current work and requires only a richer prompt template encoding the pipeline structure.

Drug Discovery and Molecular Optimization. Molecular property optimization (e.g., maximizing binding affinity while minimizing toxicity) is a canonical expensive black-box optimization problem. LLMs trained on chemical literature encode rich priors over molecular structure–activity relationships. OT-Hybrid-BO could arbitrate between LLM-proposed molecular candidates and a GP surrogate of the property surface, with the OT distance detecting when the GP's smooth surrogate is misspecified (e.g., across activity cliffs). This would require replacing the Euclidean W_2 with a Wasserstein distance defined over a molecular graph metric, a non-trivial but tractable extension.

Continual and Multi-Task BO. In continual learning settings where new tasks arrive sequentially, the LLM arm could encode cross-task transfer knowledge (e.g., "learning rate schedules that worked well on similar vision tasks") while the GP is fitted only on the current task's observations. The OT arbitration would naturally down-weight the LLM's cross-task prior as task-specific data accumulates, providing a principled mechanism for knowledge transfer without catastrophic forgetting of task-specific structure.

Open questions. Several fundamental questions remain unanswered and represent directions for future theoretical and empirical work:

- Under what conditions does $W_2(P_L, P_G)$ provide a consistent estimator of GP misspecification severity?
- Can the arbitration policy itself be meta-learned across tasks, replacing the hand-designed sigmoid schedule with a learned policy trained on historical BO traces?
- How does OT-Hybrid-BO scale to search spaces with thousands of dimensions, where both Gaussian approximations and GP surrogates are known to degrade?
- Is there a theoretical regret bound for OT-Hybrid-BO that interpolates between the LLM's prior-dependent regret and the GP's posterior-dependent regret?